

DATA STRUCTURES AND ALGORITHMS
QUESTION BANK FOR CET II Batch 2025-29

*Topics Covered: Linear Queue • Circular Queue • Double Ended Queue • Priority Queue • Singly Linked List •
Doubly Linked List • Circular Linked List*

Note: The question bank is for facilitating better preps for upcoming CET II exams.

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer all questions. Each question carries 1 Mark. Choose the correct option.

- Q1.** In a linear queue, insertion of a new element is done at the: [1 Mark]
(A) Front (B) Rear (C) Middle (D) Both ends
- Q2.** The operation used to remove an element from a queue is called: [1 Mark]
(A) Push (B) Pop (C) Dequeue (D) Peek
- Q3.** Which condition indicates queue overflow in an array-based linear queue of size n (0-indexed)? [1 Mark]
(A) $\text{front} == -1$ (B) $\text{rear} == n-1$ (C) $\text{front} == \text{rear}$ (D) $\text{rear} == 0$
- Q4.** A queue data structure follows which principle? [1 Mark]
(A) LIFO (B) FIFO (C) LILO (D) Random access
- Q5.** What is the time complexity of the enqueue operation in a simple linear queue? [1 Mark]
(A) $O(n)$ (B) $O(\log n)$ (C) $O(1)$ (D) $O(n^2)$
- Q6.** A circular queue overcomes which major limitation of a linear queue? [1 Mark]
(A) Overflow even when queue is not actually full (B) Underflow (C) Memory leak (D) Slow traversal
- Q7.** In a circular queue implemented with an array of size n , the position after rear is computed as: [1 Mark]
(A) $\text{rear} + 1$ (B) $(\text{rear} + 1) \% n$ (C) $\text{rear} - 1$ (D) $(\text{rear} - 1) \% n$
- Q8.** A circular queue is considered full when: [1 Mark]
(A) $\text{front} == \text{rear}$ (B) $(\text{rear} + 1) \% n == \text{front}$ (C) $\text{rear} == n$ (D) $\text{front} == 0$
- Q9.** Which technique makes array-based circular queue implementation efficient? [1 Mark]
(A) Recursion (B) Modulo (%) operation (C) Hashing (D) Binary search
- Q10.** A circular queue is also commonly known as a: [1 Mark]
(A) Ring buffer (B) Stack buffer (C) Linear buffer (D) Tree buffer
- Q11.** In a Deque (Double Ended Queue), insertion and deletion can be performed at: [1 Mark]
(A) Only the front (B) Only the rear (C) Both ends (D) Only the middle
- Q12.** A deque in which insertion is allowed only at one end but deletion is allowed at both ends is called: [1 Mark]
(A) Input-restricted deque (B) Output-restricted deque (C) Circular deque (D) Priority deque
- Q13.** A Deque can be used to efficiently implement: [1 Mark]
(A) Only a Stack (B) Only a Queue (C) Both a Stack and a Queue (D) Neither Stack nor Queue

- Q14.** An output-restricted deque allows deletion at: *[1 Mark]*
 (A) Both ends (B) Only one end (C) Neither end (D) The middle
- Q15.** The term "Deque" stands for: *[1 Mark]*
 (A) Double Ended Queue (B) Double End Quantity (C) Data Ended Queue (D) Dual Element Queue
- Q16.** In a priority queue, elements are removed on the basis of their: *[1 Mark]*
 (A) Insertion order (B) Priority (C) Random order (D) Size of the queue
- Q17.** Which data structure is most commonly used to implement a priority queue efficiently? *[1 Mark]*
 (A) Array (B) Linked List (C) Heap (D) Stack
- Q18.** In a max priority queue, which element is served first? *[1 Mark]*
 (A) The smallest element (B) The largest element (C) The first inserted element (D) The last inserted element
- Q19.** The time complexity of insertion in a priority queue implemented using a binary heap is: *[1 Mark]*
 (A) $O(1)$ (B) $O(n)$ (C) $O(\log n)$ (D) $O(n^2)$
- Q20.** Priority queues are commonly used in which of the following algorithms? *[1 Mark]*
 (A) Dijkstra's shortest path algorithm (B) Bubble sort (C) Linear search (D) Binary search
- Q21.** Each node in a singly linked list typically contains: *[1 Mark]*
 (A) Data and one pointer (B) Data and two pointers (C) Only data (D) Only a pointer
- Q22.** The last node of a singly linked list points to: *[1 Mark]*
 (A) The first node (B) NULL (C) Itself (D) The second node
- Q23.** The time complexity to access the n th element of a singly linked list is: *[1 Mark]*
 (A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n \log n)$
- Q24.** Which pointer is used to keep track of the beginning of a linked list? *[1 Mark]*
 (A) Tail (B) Head (C) Next (D) Current
- Q25.** Insertion of a new node at the beginning of a singly linked list takes: *[1 Mark]*
 (A) $O(n)$ (B) $O(1)$ (C) $O(\log n)$ (D) $O(n^2)$
- Q26.** Each node in a doubly linked list contains: *[1 Mark]*
 (A) Only one pointer (B) Data and two pointers (C) Only data (D) Three pointers
- Q27.** A doubly linked list allows traversal: *[1 Mark]*
 (A) Only in the forward direction (B) Only in the backward direction (C) In both forward and backward directions (D) No traversal is possible
- Q28.** Compared to a singly linked list, a doubly linked list requires: *[1 Mark]*
 (A) Less memory per node (B) More memory per node (C) The same memory per node (D) No memory
- Q29.** In a circular linked list, the last node's next pointer points to: *[1 Mark]*
 (A) NULL (B) The first node (C) Itself (D) The previous node
- Q30.** A circular linked list is particularly useful for implementing: *[1 Mark]*
 (A) Round robin CPU scheduling (B) Binary search (C) Quick sort (D) Hashing

SECTION B: VERY SHORT ANSWER QUESTIONS

Answer all questions. Each question carries 3 Marks.

- Q1.** Define a queue and state its two basic operations along with the principle it follows. [3 Marks]
- Q2.** What is a circular queue? Why is it preferred over a simple linear queue? [3 Marks]
- Q3.** Differentiate between an input-restricted deque and an output-restricted deque. [3 Marks]
- Q4.** What is a priority queue? Give one real-life example where it is used. [3 Marks]
- Q5.** What is a singly linked list? List two advantages of a linked list over an array. [3 Marks]
- Q6.** Define a doubly linked list and mention one advantage it has over a singly linked list. [3 Marks]
- Q7.** What is a circular linked list? Give one practical application of it. [3 Marks]
- Q8.** State the conditions for queue overflow and queue underflow in an array-based linear queue. [3 Marks]
- Q9.** Write the node structure of a singly linked list in C, defining data and pointer fields. [3 Marks]
- Q10.** Differentiate between a linear queue and a circular queue on the basis of memory utilization. [3 Marks]

SECTION C: SHORT ANSWER QUESTIONS

Answer all questions. Each question carries 5 Marks.

- Q1.** Explain the array-based implementation of a linear queue. Write the algorithms for the enqueue and dequeue operations. [5 Marks]
- Q2.** Explain the concept of a circular queue with a neat diagram. Write the algorithm for insertion of an element into it. [5 Marks]
- Q3.** Explain the concept of a Deque along with its two types (input-restricted and output-restricted), with suitable diagrams. [5 Marks]
- Q4.** Explain how a priority queue can be implemented using a simple array. What are the limitations of this approach? [5 Marks]
- Q5.** Write an algorithm to insert a new node at the beginning of a singly linked list, along with a suitable diagram. [5 Marks]
- Q6.** Write an algorithm to delete the last node from a doubly linked list. [5 Marks]
- Q7.** Explain the traversal process of a circular linked list and write the corresponding algorithm. [5 Marks]
- Q8.** Compare singly linked list, doubly linked list, and circular linked list on the basis of structure, memory, and traversal. [5 Marks]
- Q9.** Explain how a stack (LIFO) and a queue (FIFO) can both be implemented using a single Deque. [5 Marks]
- Q10.** Explain a min-heap based priority queue. Describe how insertion and deletion (extract-min) operations are performed on it. [5 Marks]

SECTION D: LONG ANSWER QUESTIONS

Answer any suitable number of questions as instructed. Each question carries 7 Marks.

- Q1.** Write an algorithm/program to implement a linear queue using an array, including enqueue, dequeue, and display operations, and trace it with a suitable example. [7 Marks]
- Q2.** Explain circular queue in detail with a neat diagram. Write complete algorithms for insertion and deletion operations, handling the full and empty conditions. [7 Marks]
- Q3.** Explain the implementation of a Deque using an array. Write algorithms for insertion and deletion operations at both the front and the rear ends. [7 Marks]
- Q4.** Explain the concept of a priority queue in detail. Write algorithms for insertion and deletion operations using a max-heap, and trace them with an example. [7 Marks]
- Q5.** Write algorithms for insertion of a new node into a singly linked list (a) at the beginning, (b) at the end, and (c) at a specified position. [7 Marks]
- Q6.** Write algorithms for insertion and deletion of a node at a given position in a doubly linked list. Illustrate each with a diagram. [7 Marks]
- Q7.** Write an algorithm to convert a singly linked list into a circular linked list. Also explain two real-world applications of circular linked lists. [7 Marks]
- Q8.** Explain the applications of a queue in real life and in computer science, giving at least five suitable examples with brief explanations. [7 Marks]
- Q9.** Write an algorithm to reverse a singly linked list in place (without using extra space), and trace the algorithm with a suitable example. [7 Marks]
- Q10.** Explain, with algorithm and diagram, how a circular queue can be implemented using a singly linked list instead of an array. [7 Marks]
- Q11.** Differentiate between an array and a linked list as data structures. Discuss the situations in which each should be preferred, with justification. [7 Marks]
- Q12.** Write an algorithm to detect the presence of a loop in a singly linked list and to remove it if found. Explain the logic used. [7 Marks]
- Q13.** Explain the implementation of a priority queue using a linked list. Compare its time complexity with a heap-based implementation for insertion and deletion. [7 Marks]
- Q14.** Write an algorithm to merge two sorted singly linked lists into a single sorted linked list, and trace it with a suitable example. [7 Marks]
- Q15.** Explain insertion, deletion, and traversal (both forward and backward) operations on a doubly linked list with algorithms, and state one practical application of it. [7 Marks]